

Accessible Machine Learning Classification of Triple-Negative Breast Cancer Molecular Subtypes Using Routine Clinical Data

Aja Starkey, University of Colorado Boulder, aja.starkey@colorado.edu

ABSTRACT

Triple-negative breast cancer (TNBC) comprises distinct molecular subtypes with differential treatment sensitivities, yet RNA sequencing required for subtype classification costs \$2,000-\$5,000 and remains inaccessible to most patients. This project develops machine learning classifiers to predict TNBC molecular subtypes using routinely collected clinical data. Using TCGA-BRCA data (n=175 TNBC samples) merged with Lehmann 2016 subtype labels, I systematically evaluated six diverse modeling approaches. Despite fundamental differences in architecture, all approaches converged to similar performance: macro F1-scores of 0.25-0.37, with AutoGluon's Random Forest and Neural Network achieving the highest macro F1 of ~0.37 (40% accuracy). Critically, per-subtype F1-scores remained below the clinical utility threshold of 0.65 for all subtypes (BL1: 0.47, BL2: 0.36, LAR: 0.17, M: 0.35), indicating insufficient diagnostic reliability for cancer treatment selection. These results demonstrate that routine clinical features contain meaningful but limited signal, validating the continued necessity of molecular profiling for accurate subtype-guided therapy.

1. INTRODUCTION

TNBC, defined by absence of ER, PR, and HER2 receptors, accounts for 15% of breast cancers but is responsible for disproportionate mortality due to aggressive biology and limited treatment options beyond chemotherapy. Landmark studies by Lehmann et al. [1,2] identified four stable molecular subtypes with unique therapeutic vulnerabilities: BL1 (Basal-Like 1, high proliferation, DNA damage response), BL2 (Basal-Like 2, immune infiltration, immunotherapy responsive), LAR (Luminal Androgen Receptor, AR-driven, responsive to AR antagonists), and M (Mesenchymal, stem-like features). Despite this knowledge, 85% of TNBC patients receive standard chemotherapy because RNA-seq based subtyping remains prohibitively expensive and inaccessible. This project addresses the critical gap between molecular knowledge and clinical practice by systematically evaluating

whether affordable, accessible classification is possible using routine clinical data.

2. RELATED WORK

Multiple classification systems exist for TNBC subtypes, with the Lehmann/Burstein four-subtype model showing best reproducibility [1,2,3,8]. Recent comprehensive reviews have synthesized these classification approaches and their clinical implications [8]. Subtypes demonstrate differential treatment responses in preclinical and clinical studies: LAR to AR antagonists [4], BL2 to immune checkpoint inhibitors [5]. Previous attempts at IHC-based classification using AR, CK5/6, and EGFR achieved ~65-70% concordance with RNA-seq [6], but lacked comprehensive feature sets and advanced ML approaches. Machine learning has successfully predicted cancer outcomes using clinical data [7], yet no published study has systematically explored optimal feature combinations and modern ML algorithms—including foundation models—for accessible TNBC subtyping.

3. DATA AND METHODS

3.1 Data Sources

TCGA-BRCA provides 175 TNBC samples with RNA-seq based subtype labels (ground truth from Lehmann 2016), comprehensive clinical data, and pathology reports accessed via cBioPortal [9,10]. Final merged dataset achieved 99.4% merge success (175/176 samples). Subtype distribution: BL1 n=59 (33.7%), M n=44 (25.1%), LAR n=38 (21.7%), BL2 n=34 (19.4%), yielding excellent class balance (1.74:1 imbalance ratio).

3.2 Feature Set & Processing

Thirteen routinely collected clinical features: AGE, BUFFA_HYPOXIA_SCORE, RACE, ETHNICITY, AJCC_PATHOLOGIC_TUMOR_STAGE, PATH_T_STAGE, PATH_N_STAGE, PATH_M_STAGE, AJCC_STAGING_EDITION, ICD_O_3_HISTOLOGY, RADIATION_THERAPY, PRIOR_DX. After one-hot encoding, the feature space expanded to 52 dimensions. All features available from

standard pathology reports requiring no additional molecular testing.

3.3 Modeling Approaches

I systematically evaluated: (1) Traditional ML: Logistic Regression, Random Forest, XGBoost; (2) Automated ML: AutoGluon evaluating 11 algorithms; (3) Foundation Models: TabPFN transformer. All models used stratified 80/20 train-test split (140 train, 35 test), 5-10 fold cross-validation, and hyperparameter optimization.

4. RESULTS

4.1 Model Performance

For cancer diagnosis applications, I prioritize F1-score over accuracy as it accounts for both false positives (unnecessary aggressive treatment) and false negatives (missed diagnoses).

Below summarizes performance across all approaches, with models ranked by macro F1-score—the primary metric for clinical utility.

Model Performance (Ranked by F1):

- **AutoGluon Random Forest: Macro F1 0.37 (40% acc)**
- **AutoGluon Neural Network: Macro F1 0.37 (40% acc)**
- Random Forest (baseline): Macro F1 0.340 (34.3% acc)
- AutoGluon Ensemble: Macro F1 0.340 (37.1% acc)
- XGBoost: Macro F1 0.265 (20.0% acc)
- Logistic Regression: Macro F1 0.248 (31.4% acc)
- TabPFN Transformer: Macro F1 0.219 (34.3% acc)

4.2 Per-Subtype F1-Scores

Clinical Significance: For cancer diagnosis, per-subtype F1-scores are critical as they reflect balanced performance across precision (avoiding false positives that lead to inappropriate treatment) and recall (avoiding false negatives that miss diagnoses). My target threshold was $F1 \geq 0.65$ for clinical utility, representing acceptable diagnostic reliability for treatment selection.

Best Model Performance (AutoGluon RF/NN):

- BL1: F1 0.47 (Precision 0.39, Recall 0.58) - Below threshold

- BL2: F1 0.36 (Precision 0.50, Recall 0.29) - Below threshold

- **LAR: F1 0.17 (Precision 0.20, Recall 0.14) - Critically low**

- M: F1 0.35 (Precision 0.38, Recall 0.33) - Below threshold

Critical Finding: All per-subtype F1-scores fell substantially below the 0.65 clinical utility threshold. LAR subtype performance (F1 0.17) was particularly concerning, as misclassification could lead to inappropriate androgen receptor antagonist therapy or missed treatment opportunities. The low recall for BL2 (0.29) is especially problematic as this subtype benefits from immunotherapy—missing these patients denies them potentially life-saving treatment.

4.3 Key Findings

F1-Score Performance Ceiling. Despite testing six diverse modeling approaches, macro F1-scores converged to 0.25-0.37 range, with the best models (AutoGluon RF and NN) achieving only F1 0.37—substantially below the 0.65 threshold for clinical deployment. This consistency across fundamentally different architectures suggests the ceiling reflects biological limitations rather than methodological inadequacies.

Clinical Implications of Low F1. For cancer diagnosis, F1-scores below 0.65 indicate unacceptable diagnostic unreliability. An F1 of 0.37 means that among patients receiving subtype-specific treatment recommendations, approximately 63% would receive incorrect guidance—potentially leading to inappropriate immunotherapy, missed AR-targeted therapy, or inadequate treatment intensity. Such error rates are incompatible with oncology practice standards.

Subtype-Specific Diagnostic Reliability. LAR and M subtypes showed particularly poor F1-scores (0.17 and 0.35), indicating these subtypes are essentially undetectable using clinical features alone. BL1 achieved the highest F1 (0.47) but still falls far short of diagnostic reliability. The low BL2 recall (0.29) means 71% of immunotherapy-responsive patients would be missed—a critical safety concern.

5. DISCUSSION

5.1 F1-Scores and Clinical Utility

Why F1-Score Matters for Cancer Diagnosis.

Unlike general classification tasks where accuracy suffices, cancer subtype classification demands balanced performance across precision and recall. False positives lead to inappropriate targeted therapies with significant side effects and costs (e.g., unnecessary immunotherapy causing autoimmune toxicity). False negatives result in missed treatment opportunities and suboptimal outcomes. The F1-score, as the harmonic mean of precision and recall, appropriately penalizes both error types.

Interpreting My F1 Results. The best macro F1 of 0.37 falls critically short of the 0.65 clinical utility threshold established in prior diagnostic ML studies [7]. This indicates that even the best-performing models would provide unreliable treatment guidance for the majority of patients. The particularly low LAR F1 (0.17) means AR-targeted therapy selection would be essentially random, while low BL2 recall (0.29) would deny most immunotherapy-responsive patients this potentially curative option.

Comparison to Prior Work. Previous IHC-based approaches achieving 65-70% concordance [6] likely translate to F1-scores of 0.60-0.65—substantially higher than my 0.37. This validates that protein expression markers provide clinically meaningful discrimination, while routine clinical features alone do not meet diagnostic standards. My results establish that clinical features can augment but not replace biomarker-based classification.

5.2 Clinical Implications

Insufficient for Treatment Selection. The F1-scores achieved (0.25-0.37) fall well below acceptable diagnostic reliability for oncology practice. Using these models for treatment selection would result in: (1) ~60-70% of patients receiving suboptimal therapy, (2) unnecessary toxicity from inappropriate targeted agents, (3) missed opportunities for curative immunotherapy in BL2 patients, and (4) inadequate treatment intensity for high-risk subtypes. These error rates preclude clinical deployment.

Validates Molecular Profiling Necessity. My systematic evaluation—achieving only F1 0.37 despite comprehensive modeling approaches—provides empirical validation that RNA-sequencing or targeted protein panels

remain necessary for accurate, clinically actionable subtype classification. The performance gap between my best models (F1 0.37) and IHC-based approaches (F1 ~0.60-0.65) supports investment in accessible biomarker panels rather than clinical-feature-only approaches.

Potential for Risk Stratification. While inadequate for definitive diagnosis, models achieving F1 0.37 may provide value in risk stratification or molecular testing prioritization. High-confidence predictions (where model probability >0.70) might guide which patients receive expedited molecular profiling, optimizing resource allocation in systems with limited testing capacity. However, such applications require prospective validation to ensure patient safety.

5.3 Limitations

Sample Size and F1 Stability. With only 35 test samples divided across 4 subtypes, per-subtype F1-scores have high variance (7-12 samples per subtype). LAR's F1 of 0.17 is based on just 7 test samples. Larger validation cohorts (200-500 test samples) are essential for stable F1 estimates before clinical translation.

Class Imbalance Effects. Despite relatively balanced classes overall (1.74:1 ratio), minority subtypes (LAR n=7, BL2 n=7 in test set) showed poorest F1-scores. This reflects both biological overlap and statistical limitations from small sample sizes. Future work requires balanced enrollment with ≥ 50 test samples per subtype for reliable F1 estimation.

5.4 Clinical Context: Performance vs Current Practice

Comparison to Random Baseline and Standard Care. While the achieved macro F1-score of 0.361 falls substantially below the clinical utility threshold of 0.65, it represents significant improvement over both random chance (25% accuracy, F1 approximately 0.25) and current standard practice. The best model (AutoGluon Random Forest) achieves 40% test accuracy and F1-score of 0.361—a 60% improvement in accuracy over random classification and a 44% improvement in F1-score over chance performance. This improvement is clinically meaningful when contextualized against current TNBC treatment paradigms.

Current Clinical Practice Without Molecular Testing. Presently, 85-90% of TNBC patients never receive molecular subtyping due to cost and accessibility barriers. These patients are treated with standardized chemotherapy regimens (typically anthracycline and taxane-based protocols such as AC-T, TAC, or dose-dense AC-T) regardless of molecular subtype. This one-size-fits-all approach effectively represents random assignment to subtype-appropriate therapy, with optimal treatment matching occurring by chance alone (approximately 25-30%). Response rates to standard chemotherapy show significant heterogeneity: pathological complete response rates of 40-50% overall mask substantial variation by molecular subtype, with some patients receiving suboptimal therapy for their specific molecular characteristics.

Model Performance in Clinical Context. The 40% classification accuracy and F1-score of 0.361 achieved by the AutoGluon Random Forest model represents meaningful improvement over current practice: correctly identifying 58% of BL1 patients (who require intensive chemotherapy due to high proliferation rates) compared to no stratification in standard practice; detecting 29% of BL2 patients (immunotherapy candidates) versus 0% identification without molecular testing; and providing any stratification for LAR patients (14% recall) compared to complete lack of AR-targeted therapy consideration in current protocols. While these detection rates remain insufficient for standalone diagnostic use (F1 0.361 vs threshold 0.65), they demonstrate that routine clinical features contain extractable signal about molecular subtypes that could inform clinical decision-making.

Potential for Hybrid Clinical Implementation. The modest but real improvement over random performance (F1 0.361 vs 0.25 baseline, representing 44% improvement) suggests potential value in hybrid approaches that leverage clinical features to optimize molecular testing resource allocation. High-confidence predictions (where model probability exceeds 70%) could guide immediate treatment decisions in resource-constrained settings, while low-confidence cases receive molecular confirmation testing. This tiered strategy could expand access to subtype-informed treatment

beyond the 10-15% of patients who currently receive RNA-sequencing, particularly in community oncology settings lacking comprehensive genomics infrastructure. However, such applications require rigorous prospective validation to ensure patient safety, as the current F1-score of 0.361 indicates unacceptable error rates for standalone diagnostic use.

Statistical Significance of Improvement. The 60% relative improvement in accuracy (from 25% random to 40% achieved) and 44% improvement in F1-score (from approximately 0.25 to 0.361) demonstrate that the performance ceiling is not at the random baseline. This establishes that routine clinical features do contain meaningful discriminative information about molecular subtypes, even if insufficient signal exists for diagnostic-grade classification. The consistency of this improvement across six diverse modeling approaches (all achieving F1-scores of 0.22-0.36) validates that the signal is robust and reproducible, not an artifact of specific modeling choices or overfitting to the training set.

6.1 Integration of Accessible Biomarkers

Targeted Biomarker Expansion. Clinical features alone achieve macro F1-score of 0.361 (40% test accuracy). Strategic integration of accessible biomarkers available from routine clinical testing could bridge the gap to clinical utility (F1 greater than or equal to 0.65) at substantially reduced cost compared to comprehensive RNA-sequencing. The following biomarkers represent high-value additions based on known biological associations with TNBC molecular subtypes and availability through standard clinical laboratory testing.

Critical Priority: Androgen Receptor Status. AR (androgen receptor) immunohistochemistry represents the single highest-value biomarker addition. The LAR (Luminal Androgen Receptor) subtype is defined by AR positivity, making AR status directly diagnostic for this subtype. Current models attempt to identify LAR with F1-score of 0.17 (essentially blind classification) because AR information is absent from clinical features. Adding AR status through standard IHC (cost approximately \$150-200) would likely achieve F1 greater than 0.60 for LAR subtype alone. AR testing is already routinely performed in many institutions for treatment planning with AR

antagonists (enzalutamide), making integration straightforward.

Proliferation and Tumor Markers. Ki-67 proliferation index shows strong association with BL1 subtype, which exhibits characteristically high proliferation rates (typically Ki-67 greater than 50-60%). Integration of Ki-67 could substantially improve BL1 detection beyond current 58% recall (F1 0.47). Standard breast cancer tumor markers (CA 15-3, CA 27-29, CEA) and lactate dehydrogenase (LDH, elevated in aggressive subtypes) are routinely measured in oncology practice and could provide additional discriminative signal at minimal incremental cost (combined cost approximately \$100-150 for panel).

Basal and Receptor Markers. EGFR (epidermal growth factor receptor) expression and CK5/6 (cytokeratin 5/6) are established basal markers enriched in BL1 and BL2 subtypes. Previous IHC-based classification approaches using AR, EGFR, and CK5/6 achieved approximately 65-70% concordance with RNA-seq, corresponding to estimated F1-scores of 0.60-0.65. Integration of these three markers via immunohistochemistry (combined cost approximately \$400-500) with existing clinical features and machine learning optimization may achieve F1 0.60-0.65, representing clinically meaningful performance at 20-25% of RNA-sequencing cost (\$2,000-5,000).

Immune and Inflammatory Markers. Routine complete blood count (CBC) metrics—particularly neutrophil-to-lymphocyte ratio (NLR)—correlate with tumor immune microenvironment and may distinguish BL2 subtype (characterized by immune infiltration, current F1 0.36). Systemic inflammatory markers including albumin, C-reactive protein (CRP), and erythrocyte sedimentation rate (ESR) are routinely measured and could provide additional discriminative signal at negligible incremental cost, as these tests are standard in oncology workup.

Proposed Minimal Biomarker Panel. A cost-optimized panel balancing discriminative power and accessibility would include: (1) AR status (IHC, approximately \$150-200, critical for LAR identification - current F1 0.17 would likely improve to F1 greater than 0.60), (2) Ki-67

proliferation index (IHC, approximately \$150-200, distinguishes high-proliferation BL1 - current F1 0.47), (3) EGFR and CK5/6 (IHC, approximately \$150-200 each, basal markers for BL1/BL2 distinction - current F1 0.47 and 0.36 respectively). Combined cost: approximately \$600-800 (30-40% of RNA-seq cost). Expected performance: macro F1-score of 0.60-0.65 based on prior IHC concordance studies, potentially meeting clinical utility threshold. This panel leverages existing laboratory infrastructure and requires no specialized genomics capabilities, enabling deployment in community oncology settings.

Radiomics Integration. Beyond laboratory biomarkers, radiomics features extracted from routine mammography and ultrasound imaging represent a complementary zero-incremental-cost approach. Deep learning models can quantify tumor texture, morphology, and vascularity patterns not visible to human observers. Integration of clinical features (current macro F1 0.361), radiomics features (available for all patients), and minimal biomarker panel (AR, Ki-67, EGFR, CK5/6) may achieve synergistic improvement through complementary information sources, potentially reaching F1 greater than or equal to 0.65 while maintaining costs substantially below comprehensive RNA-sequencing.

6. FUTURE WORK

6.1 Achieving Clinical F1 Thresholds

Multimodal Integration for F1 Improvement.

Clinical features alone achieve F1 0.37. Integration with accessible biomarkers (2-3 IHC markers, ~\$400) may achieve F1 0.60-0.65—approaching clinical utility. Radiomics features from routine imaging could provide complementary signal. Such hybrid approaches balance accuracy, accessibility, and cost.

Binary Classification for Higher F1. Binary tasks may achieve $F1 \geq 0.65$: (1) Basal vs Non-basal for chemotherapy selection, (2) BL2 vs Other for immunotherapy candidacy (critical given low BL2 recall), (3) LAR vs Other for AR-targeted therapy. Binary classification reduces complexity and may improve per-class F1-scores to clinically acceptable levels.

6.2 Validation with F1 Focus

External Validation Cohorts. Independent datasets (METABRIC, clinical trial cohorts) with ≥ 200 TNBC samples would provide stable per-subtype F1 estimates. Validation should report F1-scores with 95% confidence intervals, explicitly addressing whether thresholds for clinical deployment are met.

Prospective Clinical Trial. Compare treatment outcomes between: (1) standard chemotherapy, (2) model-guided therapy (if $F1 \geq 0.65$ achieved), (3) RNA-seq-guided therapy. Primary endpoint: progression-free survival. Secondary: cost-effectiveness per quality-adjusted life year. This would establish whether intermediate F1-scores (0.60-0.65) translate to acceptable clinical outcomes.

7. CONCLUSION

Through systematic evaluation of six diverse machine learning approaches, I demonstrate that routine clinical features alone achieve macro F1-scores of only 0.25-0.37 for TNBC molecular subtype classification—substantially below the 0.65 threshold required for clinical deployment. Critically, per-subtype F1-scores remained below 0.50 for all subtypes, with LAR achieving only F1 0.17 and BL2 recall of just 0.29. For cancer diagnosis where both false positives (inappropriate treatment) and false negatives (missed diagnoses) carry serious consequences, these F1-scores indicate unacceptable diagnostic unreliability.

The consistency of F1 performance (0.25-0.37) across fundamentally different modeling paradigms provides robust evidence that this ceiling reflects biological reality: molecular subtypes exhibit substantial clinical overlap despite distinct gene expression. This work establishes that routine clinical features must be augmented with accessible biomarkers to achieve $F1 \geq 0.65$ for clinically actionable subtype classification. Future work should focus on hybrid approaches combining clinical features with limited IHC panels and radiomics, aiming to achieve diagnostic reliability ($F1 \geq 0.65$) at reduced cost compared to comprehensive RNA-sequencing.

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